

Trader Behavior vs Market Sentiment - Final Report

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Objective:

The goal of this project is to analyze how trader performance (profit and trading volume) changes with overall market sentiment, specifically during Fear and Greed periods, using Bitcoin sentiment data and Hyperliquid trader activity data.

Data Used:

1. Fear and Greed Index - Shows market mood by day (Fear, Greed, etc.).
 2. Trader Data - Contains trade details such as profit, trade size, and timestamps.
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Data Preparation:

- Cleaned and formatted both datasets using Pandas.
 - Converted timestamps into date format and merged datasets on the "date" column.
 - Created new columns:
 - $\text{total_volume} = \text{execution_price} \times \text{size_tokens}$
 - $\text{profitable} = 1 \text{ if } \text{closed_pnl} > 0 \text{ else } 0$
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Key Insights:

1. Traders achieved higher profits during Greed phases.
 2. Trading activity and volume increased in Greed periods.
 3. Profit fluctuations were higher during Fear days.
 4. Market sentiment strongly influenced trader behavior and outcomes.
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Outputs Generated:

All visual charts are saved in the outputs folder:

- avg_volume_vs_sentiment.png
- profit_distribution.png
- top_10_traders.png
- correlation_heatmap.png

Processed data files are saved in the csv_files folder:

- processed_trader_data.csv
- summary_by_sentiment.csv

Tools Used:

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Conclusion:

Trading performance clearly varies with market sentiment.

During Greed phases, traders show higher confidence and earn more profits, while during Fear phases, traders act cautiously but face greater volatility.

This analysis highlights how emotions drive trading behavior and impact market outcomes.